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Network-based feedback control for systems with mixed delays based on quantization and dropout compensation[☆]Rongni Yang^{a,b,1}, Peng Shi^{b,c}, Guo-Ping Liu^{b,d}, Huijun Gao^a^a Institute of Intelligent Control and Systems, Harbin Institute of Technology, Harbin 150001, China^b Faculty of Advanced Technology, University of Glamorgan, Pontypridd, CF37 1DL, UK^c School of Engineering and Science, Victoria University, Melbourne, Vic 8001, Australia^d Center for Control Theory and Guidance Technology, Harbin Institute of Technology, Harbin 150001, China

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ABSTRACT

This paper deals with the problem of feedback control for networked systems with discrete and distributed delays subject to quantization and packet dropout. Both a state feedback controller and an observer-based output feedback controller are designed. The infinite distributed delay is introduced in the discrete networked domain for the first time. Also, it is assumed that system state or output signal is quantized before being communicated. Moreover, a compensation scheme is proposed to deal with the effect of random packet dropout through communication network. Sufficient conditions for the existence of an admissible controller are established to ensure the asymptotical stability of the resulting closed-loop system. Finally, a numerical example is given to illustrate the proposed design method in this paper.

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1. Introduction

In modern industrial systems, sensors, controllers and plants are often connected over a network medium so-called networked control systems. Due to appealing advantages in NCSs, together with fruitful applications in a broad range of areas, considerable attention has been devoted to the stability and control of NCSs; see for example, Jiang, Mao, and Shi (2010), Seiler and Sengupta (2005), Walsh, Ye, and Bushnell (2002), Yang, Xia, and Shi (2011), Yin, Yu, and Zhang (2010) and Zhao, Liu, and Rees (2009) and the references therein. Nevertheless, it is worth mentioning that the insertion of communication networks in control loops leads to some inevitable phenomena including random delay, packet dropout, quantization errors and so on, which may result in system

performance deterioration and have been primarily highlighted in the literature.

Time delays commonly exist in practical NCSs (Liu & Yang, 2011; Shi, Mahmoud, Nguang, & Ismail, 2006; Wang, Ho, Liu, & Liu, 2009; Wu, Su, Shi, & Qiu, 2011), which are of discrete nature. Another important delay, namely, distributed time delay, which has recently drawn much research interest when modeling a realistic complex system (Xie, Fridman, & Shaked, 2001), has been proposed in the NCSs for the first time in this paper. Much attention has been focused on how to deal with the distributed delays in discrete NCSs, including the construction of novel Lyapunov functions and the use of new inequality techniques. Due to the promising background in NCSs, it is meaningful to propose such a discrete term for NCSs and finally aim to apply a new term of controller, $u(k) = \sum_m \mu_m x(k-m)$, in practical NCSs.

Quantization always exists in computer-based control systems and quantization errors have adverse effects on the NCSs' performance. In early 1990s, quantized state feedback was employed to stabilize an unstable linear system by Delchamps (1990). Inspiringly, there is a new trend of research on the quantization effect on NCSs where a quantizer is regarded as an information coder. Consequently, it is necessary to conduct an analysis on the quantizers and understand how much effect the quantization makes on the overall systems.

On the other hand, due to the limited transmission capacity of the network, one of the challenging issues that has inevitably

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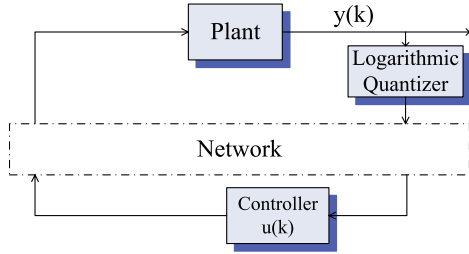


Fig. 1. The structure of NCS in (1).

emerged is data loss (Liu, 2010; Shen, Wang, & Hung, 2010; Zhao et al., 2009).

Recently, there have been three main methods to deal with control input data loss for real-time NCSs, that is to use zero control input, keep the previous one, or use the predictive control sequence (Liu, 2010). In this paper, the distributed time delays $\sum_m \mu_m x(k-m)$ have been introduced, which include all the cases mentioned above. Although in the current form the coefficients μ_m are constants, there is still potential in future that the distributed time delays can be an estimation of the output measurements that have been lost, based on the available measurements. Consequently, it is not a simple system with delays but an attempt to apply a new controller form in NCS.

In response to the above discussion, the networked-based feedback control problem for systems with discrete and distributed delays involving quantization and dropout is investigated in this paper. In the NCSs, it is assumed that the measurement signals are quantized before being communicated. The main contributions of this paper are summarized as follows. (1) The introduction of the distributed delays in the system states for NCSs and will be finally applied in controller design in future work. (2) A compensation scheme is proposed to deal with the effect of random packet dropout through communication network. (3) The practical observer-based output feedback controller is designed.

Notation: The notation used throughout the paper is fairly standard. The superscript “ T ” stands for matrix transposition. $\mathbb{R}^n$ denotes the n -dimensional Euclidean space and the notation $P > 0$ (≥ 0) means that P is real symmetric and positive definite (semi-definite). $\mathbb{E}\{\cdot\}$ denotes the expectation.

2. Problem formulation

Consider the following networked control system:

$$\begin{aligned} (\mathcal{E}): x(k+1) &= Ax(k) + Bx(k-d(k)) \\ &+ C \sum_{i=1}^{+\infty} \mu_i x(k-i) + Du(k), \\ y(k) &= Ex(k), \\ x(j) &= \phi(j), \quad j \leq 0, \end{aligned} \quad (1)$$

where $x(k) \in \mathbb{R}^n$ is the state vector, $u(k) \in \mathbb{R}^q$ is the control input, $y(k) \in \mathbb{R}^m$ is the measured system output, and A, B, C, D and E are known real matrices with appropriate dimensions. The schematic diagram for system (1) is shown in Fig. 1.

In system (1), the positive integer $d(k)$ denotes the time-varying delay satisfying $d_1 \leq d(k) \leq d_2$, $k \in \mathbb{N}^+$, where d_1 and d_2 are known positive integers. The constants $\mu_i \geq 0$ ($i = 1, 2, \dots$) satisfy the following convergence conditions:

$$\sum_{i=1}^{+\infty} i\mu_i < +\infty \quad \text{and} \quad \bar{\mu} \triangleq \sum_{i=1}^{+\infty} \mu_i < +\infty. \quad (2)$$

Remark 1. Convergence condition (2) helps to guarantee that the Lyapunov function V_5 in (13) is convergent, and deal with

the distributed delay. Furthermore, the form proposed here, $(\sum_i \mu_i x(k-i))$, can be directly used in controller design for NCS based on the networked predictive method (Liu, 2010; Zhao et al., 2009) in our subsequent work.

Note that the measured signals will be quantized through the network before they are transmitted to the controller. The set of quantized levels is described as $\mathcal{U} = \{\pm u_i, u_i = \rho^i u_0, i = 0, \pm 1, \pm 2, \dots\} \cup \{0\}$, $0 < \rho < 1$, $u_0 > 0$, where the parameter ρ is called the quantization density, and the logarithmic quantizer $q(\cdot)$ is defined as

$$q(v) = \begin{cases} u_i, & \text{if } u_i < v \leq \bar{u}_i, \\ 0, & \text{if } v = 0, \\ -q(-v), & \text{if } v < 0. \end{cases} \quad (3)$$

with $\delta = (1-\rho)/(1+\rho)$, $\underline{u}_i = \frac{1}{1+\delta}u_i$, $\bar{u}_i = \frac{1}{1-\delta}u_i$.

The controllers based on the quantized signals will be designed such that the desired dynamic performance of system (1) is achieved while the data packet dropout arises.

Before ending this section, let us recall the following results which will be used in the next section.

Lemma 1. Let $M \in \mathbb{R}^{n \times n}$ be a positive semi-definite matrix, $x_i \in \mathbb{R}^n$ and $a_i \geq 0$ ($i = 1, 2, \dots$). If the series concerned are convergent, the following inequality holds:

$$\left(\sum_{i=1}^{+\infty} a_i x_i \right)^T M \left(\sum_{i=1}^{+\infty} a_i x_i \right) \leq \left(\sum_{i=1}^{+\infty} a_i \right) \sum_{i=1}^{+\infty} a_i x_i^T M x_i. \quad (4)$$

Lemma 2. For any vectors $x, y \in \mathbb{R}^n$, matrices D, E, F with appropriate dimensions, and any scalar $\varepsilon > 0$, if $F^T F \leq I$, then

$$DFE + E^T F^T D^T \leq \varepsilon DD^T + \varepsilon^{-1} E^T E. \quad (5)$$

3. State feedback control

Assume that the state of system (1) is measurable and will be quantized before it is transmitted to the controller through communication network while the data packet dropout happens. To realize the state feedback control for system (1) subject to quantization and packet dropout, the following compensation scheme is constructed at the side of the controller in order to cope with the effect of data loss:

$$\hat{x}(k) = (1 - \theta_k) q(x(k)) + \theta_k \hat{x}(k-1), \quad (6)$$

where $q(\cdot)$ is the logarithmic quantizer defined in (3). Then, a state feedback controller based on quantized state information and packet dropout compensation is designed as

$$u(k) = K\hat{x}(k), \quad (7)$$

where the matrix K will be designed later. Thus, the resulting closed-loop system is as follows

$$\begin{aligned} x(k+1) &= Ax(k) + Bx(k-d(k)) + C \sum_{i=1}^{+\infty} \mu_i x(k-i) \\ &+ (1 - \theta_k) DKq(x(k)) + \theta_k DK\hat{x}(k-1). \end{aligned} \quad (8)$$

Then the quantizing effects can be transformed into sector bounded uncertainties:

$$q(x(k)) - x(k) = \Delta_k x(k), \quad \|\Delta_k\| \leq \delta, \quad (9)$$

for a given quantization density ρ . Setting $\xi^T(k) = [x^T(k) \hat{x}^T(k-1)]$, the combination of (6) and (8) yields the following augmented system

$$\xi(k+1) = (\bar{A} + \Delta\bar{A} + \bar{D} + \Delta\bar{D})\xi(k) + \bar{B}x(k-d(k))$$

$$+ \bar{C} \sum_{i=1}^{+\infty} \mu_i x(k-i), \quad (10)$$

where

$$\begin{aligned}\bar{A} &= \begin{bmatrix} A + \hat{\theta}DK & \bar{\theta}DK \\ \hat{\theta}I & \bar{\theta}I \end{bmatrix}, \quad \bar{D} = \begin{bmatrix} \tilde{\theta}DK & -\tilde{\theta}DK \\ \tilde{\theta}I & -\tilde{\theta}I \end{bmatrix}, \\ \Delta\bar{A} &= \begin{bmatrix} \hat{\theta}DK \\ \hat{\theta}I \end{bmatrix} \Delta_k \begin{bmatrix} I & 0 \end{bmatrix}, \\ \Delta\bar{D} &= \begin{bmatrix} \tilde{\theta}DK \\ \tilde{\theta}I \end{bmatrix} \Delta_k \begin{bmatrix} I & 0 \end{bmatrix}, \\ \bar{B}^T &= [B^T \quad 0], \\ \bar{C}^T &= [C^T \quad 0], \quad \hat{\theta} = 1 - \bar{\theta}, \quad \tilde{\theta} = \bar{\theta} - \theta_k.\end{aligned}$$

Theorem 1. Consider the discrete NCS in (1). For a given quantization density $\rho > 0$ and packet-loss rate $\bar{\theta}$, closed system (10) with controller (7) is asymptotically stable in the mean square if there exist matrices $P > 0$, $Q > 0$, $R > 0$, $S > 0$, $M > 0$, $N > 0$, K and scalar $\varepsilon > 0$ satisfying:

$$\Phi = \begin{bmatrix} \phi_{11} + \tilde{\phi}_{11} & * & * \\ \phi_{21} & \phi_{22} & * \\ 0 & \phi_{32} & -\varepsilon I \end{bmatrix} < 0, \quad (11)$$

$$PM = I, \quad QN = I, \quad (12)$$

where

$$\begin{aligned}\phi_{11} &= \text{diag} \left\{ -P + (d_2 - d_1 + 1)R + \bar{\mu}S, -Q, -R, -\frac{1}{\bar{\mu}}S \right\}, \\ \tilde{\phi}_{11} &= \text{diag} \{ \delta^2 \varepsilon, 0, 0, 0 \}, \\ \phi_{21} &= \begin{bmatrix} A + \hat{\theta}DK & \bar{\theta}DK & B & C \\ DK & -DK & 0 & 0 \\ \hat{\theta}I & \bar{\theta}I & 0 & 0 \\ I & -I & 0 & 0 \end{bmatrix}, \\ \phi_{22} &= \text{diag} \left\{ -M, -\sigma_{\bar{\theta}}^{-1}M, -N, -\sigma_{\bar{\theta}}^{-1}N \right\}, \\ \phi_{32} &= [\hat{\theta}K^T D^T \quad K^T D^T \quad \hat{\theta}I \quad I].\end{aligned}$$

Proof. Construct the following Lyapunov–Krasovskii function:

$$\begin{aligned}V(k) &= V_1(k) + V_2(k) + V_3(k) + V_4(k) + V_5(k), \\ V_1(k) &= x^T(k)Px(k), \quad V_2(k) = \hat{x}^T(k-1)Q\hat{x}(k-1), \\ V_3(k) &= \sum_{i=k-d(k)}^{k-1} x^T(i)Rx(i), \\ V_4(k) &= \sum_{j=k-d_2+1}^{k-d_1} \sum_{i=j}^{k-1} x^T(i)Rx(i), \\ V_5(k) &= \sum_{i=1}^{+\infty} \mu_i \sum_{j=k-i}^{k-1} x^T(j)Sx(j).\end{aligned} \quad (13)$$

Calculating the difference of $V(k)$ along system (10) and taking the mathematical expectation $\mathbb{E}\{\Delta V_i(k)\} = \mathbb{E}\{V_i(k+1)|k\} - V_i(k)$, we have

$$\begin{aligned}\mathbb{E}\{\Delta V_1(k)\} &= W_{P_1}^T P W_{P_1} + \sigma_{\bar{\theta}} W_{P_2}^T P W_{P_2} - x^T(k)Px(k), \\ \mathbb{E}\{\Delta V_2(k)\} &= W_{Q_1}^T Q W_{Q_1} + \sigma_{\bar{\theta}} W_{Q_2}^T Q W_{Q_2} \\ &\quad - \hat{x}^T(k-1)Q\hat{x}(k-1), \\ \mathbb{E}\{\Delta V_3(k)\} &\leq x^T(k)Rx(k) - x^T(k-d(k))Rx(k-d(k)) \\ &\quad + \sum_{j=k-d_2+1}^{k-d_1} x^T(j)Rx(j),\end{aligned} \quad (14)$$

$$\begin{aligned}\mathbb{E}\{\Delta V_4(k)\} &= (d_2 - d_1) x^T(k)Rx(k) - \sum_{j=k-d_2+1}^{k-d_1} x^T(j)Rx(j), \\ \mathbb{E}\{\Delta V_5(k)\} &\leq \bar{\mu} x^T(k)Sx(k) \\ &\quad - \frac{1}{\bar{\mu}} \left(\sum_{i=1}^{+\infty} \mu_i x(k-i) \right)^T S \left(\sum_{i=1}^{+\infty} \mu_i x(k-i) \right)\end{aligned}$$

with

$$\begin{aligned}W_{P_1} &= (A + \hat{\theta}DK)x(k) + \bar{\theta}DK\hat{x}(k-1) + \hat{\theta} \\ &\quad \times DK\Delta_k x(k) + Bx(k-d(k)) + C \sum_{i=1}^{+\infty} \mu_i x(k-i), \\ W_{P_2} &= DKx(k) - DK\hat{x}(k-1) + DK\Delta_k x(k), \\ W_{Q_1} &= \hat{\theta}x(k) + \bar{\theta}\hat{x}(k-1) + \hat{\theta}\Delta_k x(k), \\ W_{Q_2} &= x(k) - \hat{x}(k-1) + \Delta_k x(k).\end{aligned}$$

A combination of (14) leads to

$$\mathbb{E}\{\Delta V(k)\} \leq \eta^T(k) \bar{\Phi} \eta(k),$$

where

$$\eta(k) = \begin{bmatrix} \xi^T(k) & x^T(k-d(k)) & \sum_{i=1}^{+\infty} \mu_i x^T(k-i) \end{bmatrix}^T,$$

$$\bar{\Phi} = \phi_{11} - (\phi_{21} + \bar{\phi}_{21})^T \bar{\phi}_{22}^{-1} (\phi_{21} + \bar{\phi}_{21}),$$

$$\bar{\phi}_{21} = [\bar{\phi}_{21,1} \quad 0],$$

$$\bar{\phi}_{21,1}^T = \Delta_k^T [\hat{\theta}K^T D^T \quad K^T D^T \quad \hat{\theta}I \quad I].$$

According to Schur Lemma, $\bar{\Phi} < 0$ is equivalent to $\tilde{\Phi} < 0$ and $\Phi < 0$ is equivalent to $\hat{\Phi} < 0$, and

$$\tilde{\Phi} = \begin{bmatrix} \phi_{11} & * \\ \phi_{21} + \bar{\phi}_{21} & \bar{\phi}_{22} \end{bmatrix},$$

$$\hat{\Phi} = \begin{bmatrix} \phi_{11} + \tilde{\phi}_{11} & * \\ \phi_{21} & \phi_{22} \end{bmatrix} + \varepsilon^{-1} \begin{bmatrix} 0 \\ \phi_{32}^T \end{bmatrix} \begin{bmatrix} 0 & \phi_{32} \end{bmatrix},$$

where $\tilde{\phi}_{22} = \text{diag} \{ -P^{-1}, -\sigma_{\bar{\theta}}^{-1}P^{-1}, -Q^{-1}, -\sigma_{\bar{\theta}}^{-1}Q^{-1} \}$. Let $M = P^{-1}$ and $N = Q^{-1}$; according to Lemma 2, we have $\tilde{\Phi} \leq \hat{\Phi}$. Hence, by (11) in Theorem 1, we readily obtain $\bar{\Phi} < 0$, which means $\mathbb{E}\{\Delta V(k)\} \leq 0$. From Lyapunov stability theory, we can conclude that closed system (10) with controller (7) is asymptotically stable in the mean square, which completes the proof of Theorem 1. $\square$

Remark 2. The addressed formulation in this paper is based on the real-world application of networked control systems and the chosen modeling framework (1) originates from the basic NCSs model $x(k+1) = Ax(k) + Bu(k)$; see Refs. Liu (2010) and Zhao et al. (2009). The merit of the proposed results lies in that the introduction of the distributed time delays $\sum_m \mu_m x(k-m)$ and much attention has been focused on how to deal with the distributed delays in discrete NCSs, including the construction of novel Lyapunov functions and new inequality techniques. Consequently, it is not a simple system with delays. We finally aim to develop a completely new term of the controller $u(k) = \sum_m \mu_m x(k-m)$.

Remark 3. It should be noted that the obtained conditions in Theorem 1 are not strict LMI conditions due to (12). However, we can solve this problem by converting it into a sequential optimization problem subject to LMI constraints.

By employing a cone complementary linearization approach, the following minimization problem based on LMI conditions is suggested.

Problem SFC (state feedback control)
 $\min \text{tr} (PM + QN)$ subject to (11) and (15)

$$\begin{bmatrix} P & I \\ I & M \end{bmatrix} \geq 0, \quad \begin{bmatrix} Q & I \\ I & N \end{bmatrix} \geq 0. \quad (15)$$

The above minimization problem can be solved through the iterative algorithm.

4. Observer based output feedback control

Suppose that the system output is available and will be quantized before it is transmitted to the controller through network. Similarly, the following compensator is constructed to deal with the packet dropout:

$$y_c(k) = (1 - \theta_k) q(y(k)) + \theta_k y_c(k-1), \quad (16)$$

where $q(\cdot)$ is the logarithmic quantizer defined in (3). Then, the following output feedback controller is constructed:

$$\begin{aligned} \tilde{x}(k+1) &= A\tilde{x}(k) + Du(k) + L(y_c(k) - E\tilde{x}(k)), \\ u(k) &= K\tilde{x}(k), \end{aligned} \quad (17)$$

where the gain matrices L and K will be designed later.

As mentioned above, the quantization error can be treated as sector bounded uncertainties as follows:

$$q(y(k)) - y(k) = \Delta_k y(k), \quad \|\Delta_k\| \leq \delta,$$

for a given quantization density ρ . Define $e(k) = x(k) - \tilde{x}(k)$, and we obtain the following augmented error system:

$$\begin{aligned} e(k+1) &= (A - LE)e(k) + Bx(k-d(k)) + C \sum_{i=1}^{+\infty} \mu_i x(k-i) \\ &\quad + [\theta_k LE - (1 - \theta_k) L \Delta_k E] x(k) - \theta_k L y_c(k-1). \end{aligned}$$

Further, denoting $\zeta^T(t) = [x^T(k) \quad e^T(k) \quad y_c^T(k-1)]^T$, we obtain the error dynamic system as follows:

$$\begin{aligned} \zeta(k+1) &= (\bar{A} + \Delta\bar{A} + \bar{D} + \Delta\bar{D}) \zeta(k) \\ &\quad + \bar{B}x(k-d(k)) + \bar{C} \sum_{i=1}^{+\infty} \mu_i x(k-i), \end{aligned} \quad (18)$$

with

$$\bar{A} = \begin{bmatrix} A + DK & -DK & 0 \\ \bar{\theta}LE & A - LE & -\bar{\theta}L \\ \hat{\theta}E & 0 & \bar{\theta}I \end{bmatrix},$$

$$\bar{D} = \begin{bmatrix} 0 & 0 & 0 \\ -\bar{\theta}LE & 0 & \bar{\theta}L \\ \bar{\theta}E & 0 & -\bar{\theta}I \end{bmatrix},$$

$$\Delta\bar{A} = \begin{bmatrix} 0 \\ -\hat{\theta}L \\ \hat{\theta}I \end{bmatrix} [\Delta_k E \quad 0 \quad 0],$$

$$\Delta\bar{D} = \begin{bmatrix} 0 \\ -\tilde{\theta}L \\ \tilde{\theta}I \end{bmatrix} [\Delta_k E \quad 0 \quad 0],$$

$$\bar{B}^T = [B^T \quad B^T \quad 0], \quad \bar{C}^T = [C^T \quad C^T \quad 0].$$

Next, let us focus our attention on the observer based output feedback control design problem for system (1), which is summarized in the following theorem.

Theorem 2. For a given quantization density $\rho > 0$ and packet-loss rate $\bar{\theta}$, closed-loop system (18) is asymptotically stable in the mean square, if there exist matrices $P > 0$, $Q > 0$, $R > 0$, $S > 0$, $Z > 0$, $M > 0$, $N > 0$, $X > 0$, L , K and scalar $\varepsilon > 0$ satisfying:

$$\Psi = \begin{bmatrix} \psi_{11} + \tilde{\psi}_{11} & * & * \\ \psi_{21} & \psi_{22} & * \\ 0 & \psi_{32} & -\varepsilon I \end{bmatrix} < 0, \quad (19)$$

$$PM = I, \quad QN = I, \quad XZ = I,$$

where

$$\psi_{11} = \text{diag} \left\{ -P + (d_2 - d_1 + 1)R + \bar{\mu}S, -Q, -Z, -R, -\frac{1}{\bar{\mu}}S \right\},$$

$$\tilde{\psi}_{11} = \text{diag} \{ \delta^2 \varepsilon E^T E, 0, 0, 0, 0 \},$$

$$\psi_{22} = \text{diag} \{ -M, -N, -X, -\sigma_{\bar{\theta}}^{-1}M, -\sigma_{\bar{\theta}}^{-1}X \},$$

$$\psi_{32} = [0 \quad -\hat{\theta}L^T \quad \hat{\theta}I \quad L^T \quad I],$$

$$\psi_{21} = \begin{bmatrix} A + DK & -DK & 0 & B & C \\ \bar{\theta}LE & A - LE & -\bar{\theta}L & B & C \\ \hat{\theta}E & 0 & \bar{\theta}I & 0 & 0 \\ LE & 0 & -L & 0 & 0 \\ E & 0 & -I & 0 & 0 \end{bmatrix}.$$

Proof. Choose the Lyapunov function candidate as

$$V(k) = V_1(k) + V_2(k) + V_3(k) + V_4(k),$$

$$V_1(k) = x^T(k)Px(k) + e^T(k)Qe(k) + y_c^T(k-1)Zy_c(k-1),$$

$$V_2(k) = \sum_{i=k-d(k)}^{k-1} x^T(i)Rx(i),$$

$$V_3(k) = \sum_{j=k-d_2+1}^{k-d_1} \sum_{i=j}^{k-1} x^T(i)Rx(i),$$

$$V_4(k) = \sum_{i=1}^{+\infty} \mu_i \sum_{j=k-i}^{k-1} x^T(j)Sx(j).$$

Thus, according to a similar proof of Theorem 1, it follows from (18) that

$$\mathbb{E} \{ \Delta V(k) \} = \mathbb{E} \{ V(k+1) | k \} - V(k) \leq \eta^T(t) \bar{\Psi} \eta(t),$$

$$\bar{\Psi} = \psi_{11} - (\psi_{21} + \tilde{\psi}_{21})^T \psi_{22}^{-1} (\psi_{21} + \tilde{\psi}_{21}),$$

$$\hat{\phi}_{21} = [\hat{\phi}_{21,1} \quad 0],$$

$$\hat{\phi}_{21,1}^T = E^T \Delta_k^T [0 \quad -\hat{\theta}L^T \quad \hat{\theta}I \quad L^T \quad I].$$

According to Schur Lemma and Lemma 2, together with (19), it can be shown that $\bar{\Psi} < 0$, which means the closed loop error dynamic system (18) is asymptotically stable in the mean square. The proof of Theorem 2 is thus completed. $\square$

5. Numerical example

In this section, a numerical example is presented to illustrate the effectiveness of the design method obtained in the previous section. Consider system (E) with the following parameters:

$$A = \begin{bmatrix} 0.6 & -0.1 & 0 \\ 0 & -0.8 & 0.5 \\ 0.2 & 0 & -0.7 \end{bmatrix}, \quad B = \begin{bmatrix} 0.2 & -0.1 & 0 \\ 0.1 & -0.1 & 0 \\ 0 & -0.2 & -0.1 \end{bmatrix},$$

$$C = \begin{bmatrix} -0.2 & 0 & 0.1 \\ -0.2 & -0.1 & 0.1 \\ 0 & 0.2 & -0.1 \end{bmatrix}, \quad D^T = [0.1 \quad 0.2 \quad -2],$$

$$E = [1 \quad 0.2 \quad 0.6],$$

$$d(k) = 2 + \frac{1 + (-1)^k}{2}, \quad \mu_i = 2^{-(i+1)}.$$

Assume that the random variable θ_k satisfy $\bar{\theta} = \mathbb{E} \{ \theta_k \} = 0.8$, $\sigma_{\bar{\theta}} = \mathbb{E} \{ (\theta_k - \bar{\theta})^2 \} = 0.16$. It is easy to verify that $d_1 = 2$, $d_2 = 3$,

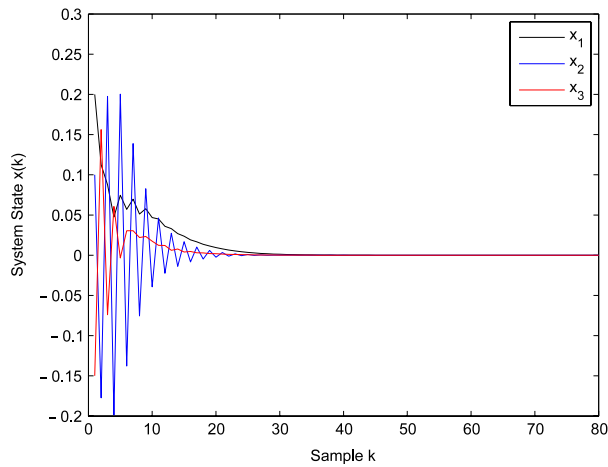


Fig. 2. State trajectories of $x(k)$.

$\bar{\mu} = 1/2$. Suppose that the above system states are unavailable. Thus, an observer based output feedback controller is designed to guarantee stability of the addressed NCS.

For a given quantization density $\rho = 0.6$ and the above parameters, by solving (19), then the desired controller in (17) can be designed with

$$K = [0.0014 \quad -0.0009 \quad -0.0034],$$

$$L^T = [0.0334 \quad -0.0828 \quad -0.0004].$$

The result is further verified by the simulation results given by Fig. 2. The initial condition is set to be $x(0) = [0.2 \quad 0.1 \quad -0.15]$. Fig. 2 shows that the system state $x(k)$ converges to zero.

6. Conclusions

In this paper, the feedback control problem for networked systems with discrete and infinite distributed delays subject to quantization and packet dropout has been studied. Specially, the networked system under investigation includes the term of the infinite distributed time delay. Also, it is assumed that the system

state or output signal is quantized before being communicated. Moreover, a compensation scheme is proposed to cope with the effect of random packet dropout in the communication network. A simulation example is exploited to illustrate the effectiveness of the proposed design procedures.

Based on the results obtained in this paper, a new form of controller $u(k) = \sum_i k_i x(k-i)$ will be developed and most efforts will be made to solve different k_i in our subsequent work.

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